

3. (a) (i) Complete the following sentences about sense organs using some of the words from the list below. [2]

impulses signs stimuli messages signals

Sense organs respond to specific by sending
information in the form of which are carried to the central
nervous system.

- (ii) State the name of:
- I. the specialised nerve cells that carry the information to the central nervous system; [1]
.....
 - II. an organ that processes information received from sense organs. [1]
.....

(b) Jed investigated the hearing of some year 11 pupils and their teachers.
He asked 12 pupils and 12 teachers to say if they could hear a buzzer when he sounded it at each of five distances away from the two groups. The same buzzer was pressed for each person.

The results are shown in the table.

Distance from buzzer (m)	Number hearing buzzer	
	Year 11 pupils	Teachers
5	12	12
10	12	12
15	12	12
20	11	7
25	8	2



Examiner
only

(i) Give **one** conclusion that Jed can make from this investigation. [1]

.....

.....

.....

.....

(ii) State **one** way in which Jed's investigation is a fair test. [1]

.....

(iii) What else should Jed have done to make sure the investigation was a fair test? [1]

.....

(iv) Jed thought that increasing the sample size would improve his confidence in his conclusion.

Describe how Jed would increase the sample size in this investigation. Explain why it would improve his confidence in his conclusion. [2]

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